

Programming Assignment Written Questions

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1) Heuristic Design & Admissibility

a. steps

- $h(n)$ = # of different letters from n & goal
- at minimum, the cost equals the length of both strings, if every letter does not match
- because each step is one, it cannot go over the lowest cost from n -to-goal

b. scrabble cost

- $h(n)$ = # of different letters, with the scrabble costs coming from the goal letters

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↳ for i in range(len(n)):  
    if n[i] != goal[i]:  
        h = h + scrabble_cost[goal[i]]
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- for every mismatch, it must at some point match the goal. those goal letters, and their scrabble costs, cannot be avoided
- this cannot overestimate because it does not take into account other scrabble costs that aren't necessarily in the goal

c. frequency

- $h(n)$ = mismatched letters + goal rarity
- the rarity of the goal word is unavoidable
- this cannot overestimate due to only accounting for the goal word, and not the intermediate words.

2)

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
WHY → ASK	Steps	5	773	16	5	97.93%
WHY → ASK	Scrabble	9	765	9	13	98.30%
WHY → ASK	Frequency	38.85	808	38.85	777	3.83%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2981	11	6	99.80%
BEAR → DUCK	Frequency	38.24	1084	38.24	788	17.31%

1. they always gave the same optimal cost
2. for both WHY → ASK & BEAR → DUCK, the scrabble cost gave the highest reduction
3. the frequency cost measure was the least informative.

4.

- steps cost always returns 1, so UCS has no "cheapest cost". the A* search has an accurate heuristic for the remaining cost, expanding fewer nodes.
- the scrabble heuristic is more useful as each letter doesn't have the same cost.
- the frequency heuristic is the least useful. it only takes into account the rarity of the goal word, leading to underestimation and more nodes expanded.

3)

1. steps

- direction independent (going to/from has cost of one)
- if reversing is always the same cost, $W1 \neq W2$ must equal optimal cost from $W2$ to $W1$

2. scramble

- direction dependent
- dog $\rightarrow$ cog: cost of 3
- cog $\rightarrow$ dog: cost of 2
- $W1 \rightarrow W2 \neq W2 \rightarrow W1$

3. frequency

- direction dependent
- dog $\rightarrow$ cog $\neq$ cog $\rightarrow$ dog might not have the same rarity
- $\therefore W2 \rightarrow W1 \neq W1 \rightarrow W2$. might be, or might not.

4)

1.

Gemini 3.1 Pro

Prompt used was specified in the instructions

Let g denote the goal state and $r(w)$ denote the nonnegative rarity of a word w . The proposed rarity-aware heuristic $h(n)$ is defined as:

$$h(n) = \begin{cases} 0 & \text{if } n = g \\ h_0(n) + r(g) & \text{if } n \neq g \end{cases}$$

Admissibility

A heuristic is admissible if it never overestimates the true minimum path cost to the goal, $h^*(n)$.

For any starting state $n \neq g$, any valid sequence of actions to reach the goal produces a path of states $n = w_0, w_1, w_2, \dots, w_k = g$.

Because every legal action changes exactly one letter, the minimum number of transitions k required to reach the goal is bounded below by the number of differing letters (the Hamming distance). Therefore, $k \geq h_0(n)$.

The actual cost of traversing this path is the sum of the costs of entering each subsequent state:

$$\text{Cost} = \sum_{i=1}^k (1 + r(w_i)) = k + \sum_{i=1}^k r(w_i)$$

Because the sequence must ultimately terminate at the goal state, w_k is always g . We can pull the guaranteed cost of entering the goal state out of the summation:

$$\text{Cost} = k + r(g) + \sum_{i=1}^{k-1} r(w_i)$$

Since the problem defines all rarity values as nonnegative ($r(w) \geq 0$), the sum of the rarities for all intermediate states must be ≥ 0 . Substituting the lower bound of k ($k \geq h_0(n)$) gives:

$$\text{Cost} \geq h_0(n) + r(g) + 0$$

Since this holds true for any valid path to the goal, it must hold true for the optimal path:

$$h^*(n) \geq h_0(n) + r(g)$$

Thus, $h(n) \leq h^*(n)$ for all n , proving the heuristic remains strictly admissible.

Informativeness

A heuristic is more informative if it yields higher values without violating admissibility, which reduces the search space by expanding fewer nodes.

By definition, $h_0(n) + r(g) \geq h_0(n)$ because $r(g)$ is nonnegative, ensuring the proposed heuristic is always at least as large as the baseline. For any game instance where the designated goal word has a rarity greater than zero ($r(g) > 0$), $h(n)$ evaluates to a strictly larger number than $h_0(n)$ for every state except the goal itself.

This enhancement achieves strict improvement in informativeness at virtually zero computational cost, as $r(g)$ only needs to be looked up in the dictionary once at the beginning of the search, rather than evaluated dynamically for every generated node.

3. it is admissible. it says the max # of transmissions k require at least k actions. the final cost is at least $k + r(g)$, the heuristic value, so it can't overestimate.

4. I actually incorrectly implemented the frequency heuristic. For the Frequency A* search, the AI version expands fewer nodes, (WHE $\rightarrow$ ASK: 634, and BEAR $\rightarrow$ DUCK: 302), giving reductions of 14.4% & 61.7% reduction respectively.

The solution costs remain the same. The AI solution has a higher reduction due to the correct implementation of the heuristic.